Name: ____________________________ Date: ________________ Period: _________

Problems 1–6

1. Solve each equation for all solutions (general solution). Express answers in exact form using π.

(a) $\sin\theta = \dfrac{\sqrt{2}}{2}$

$\theta =$ ______________________________

(b) $\cos\theta = -\dfrac{1}{2}$

$\theta =$ ______________________________

(c) $\tan\theta = -1$

$\theta =$ ______________________________

2. Solve $2\sin\theta - 1 = 0$ for all θ in $[0, 2\pi]$. Show each step.

Step 1 — Isolate $\sin\theta$: ______________________________

Step 2 — Reference angle: ______________________________

Step 3 — Identify quadrants and write solutions: ______________________________

____

3. Solve $2\cos^2\theta - \cos\theta - 1 = 0$ for all θ in $[0, 2\pi]$. (*Hint: treat* $\cos\theta$ *as the variable and factor.*)

Let $u = \cos\theta$. Factor: ______________________________

Case 1: $u =$ ______ $\Rightarrow \theta =$ ______________________________

Case 2: $u =$ ______ $\Rightarrow \theta =$ ______________________________

4. **(Contextual)** A model for the height (in meters) of a Ferris wheel car is $h(t) = 8\sin\left(\dfrac{\pi}{15}t\right) + 10$, where t is time in seconds after the ride starts, $0 \le t \le 60$.

(a) For what values of t in $[0, 60]$ is $h(t) = 14$? Show your work.

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(b) For what values of t in $[0, 60]$ is $h(t) \ge 14$? Write your answer as an interval or union of intervals.

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5. **(AP-Style MC)** Which of the following gives all solutions to $\cos\theta = \dfrac{1}{2}$ for $\theta \in [0, 2\pi]$?

(A) $\theta = \dfrac{\pi}{3}$ only

(B) $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$

(C) $\theta = \frac{\pi}{3} + 2\pi k$ for integer k

(D) $\theta = \frac{\pi}{6}$ and $\theta = \frac{5\pi}{6}$

(E) $\theta = \frac{\pi}{3}$ and $\theta = \frac{2\pi}{3}$

Answer: ________

6. (AP-Style MC) A periodic function f satisfies $f(\theta) = 0.6$ for exactly two values of θ in $(0, 2\pi)$. Which of the following could be f?

(A) $f(\theta) = \cos\theta$

(B) $f(\theta) = \sin\theta$

(C) $f(\theta) = \tan\theta$

(D) $f(\theta) = \sin^2\theta$

(E) Both (A) and (B)

Answer: ________

Extension optional

Extension (Optional): Solve $2\sin^2\theta - \sin\theta - 1 = 0$ for all real θ.

(a) Let $u = \sin\theta$ and factor the quadratic in u.

(b) Solve each resulting equation for θ and write the general solution.

(c) How many solutions are in the interval $[0, 4\pi]$? List them. ________

Preview — Lesson 3.11: Secant, Cosecant, and Cotangent
You have now solved equations involving sine, cosine, and tangent. In Lesson 3.11 we meet three reciprocal functions — secant, cosecant, and cotangent — and explore how their graphs and asymptotic behavior arise directly from the functions you already know.